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The Orchestration Problem: An Out-of-Path AI Supervisor for Governance-Constrained Tactical-Edge Substrate

Justin H. Kuiper, CISSP

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Abstract

P3 established DTN custody transfer as the transport primitive for moving state across variable links. P4 established the WAL + CRDT architecture as the state coherence primitive for ensuring that moved state converges to a coherent operational picture. What neither paper addresses is how the decisions required to operate these substrates — which bundles to prioritize, when to initiate custody transfer, which merges to escalate, how to respond when a contact window opens unexpectedly — are made at operational tempo, by a node that may have no meaningful connectivity to human authority for hours or days.

This paper argues that the orchestration of the tactical substrate requires an out-of-path AI Supervisor: an AI system that monitors the substrate's state, reasons over the governance policies encoded by the H half, and makes operational decisions within a tightly-bounded autonomy envelope without sitting on the critical path of any individual custody transfer or state merge. The critical constraint is "out-of-path": an orchestrator that sits on the critical path of every operation introduces a single point of failure, creates a bottleneck that degrades under load, and produces a cascade failure when it fails or disconnects. An out-of-path Supervisor fails gracefully: if the Supervisor is unavailable, the substrate continues to operate under its last-set governance parameters until the Supervisor recovers.

The paper develops the architectural argument for the out-of-path design, specifies what the Supervisor must be able to observe (the WAL's event stream and the CRDT state index), what it must be able to decide (within its autonomy envelope), what it must escalate (at the governance boundary), and how its model is maintained under DDIL conditions when the Supervisor cannot be updated through conventional model management channels. The HGC^3AE^2 framework is the governing architecture: the Supervisor is not the governance authority but the executor of governance policy, operating within the authority envelope that the H half defines, logging its decisions as WAL entries that the AE^2 mechanism makes auditable.

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ABSTRACT

The orchestration function in the Mission-Ready operating model (MR-P6) is in-path: the Orchestrator routes work, monitors execution, and surfaces escalation signals from a position embedded within the work flow. On the tactical substrate (EDO-P1), the in-path Orchestrator is structurally fragile — partition breaks its visibility, capacity degradation degrades its responsiveness, adversarial pressure attacks its decision logic. This paper specifies an **out-of-path AI supervisor**: a supervision layer that observes the in-path Orchestrator's decisions and signals anomaly, drift, or envelope breach without being itself in the work-flow path. The supervisor is monitor, not monitoree. It can fail without halting work; it can be partitioned from the Orchestrator without losing its supervision premise; it can be attacked without bringing down the operating model. The argument is from safety-critical systems engineering: monitor and monitoree should not share a failure surface.[1][2]

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1. INTRODUCTION

The Operating Model paper (MR-P6) places the Orchestrator function in-path: it routes work, monitors completion, surfaces escalations. The in-path placement is operationally efficient — the Orchestrator sees the work as it flows — and is well-suited to cloud-native conditions where the substrate supports continuous observation.[3] On the tactical substrate (EDO-P1), the in-path Orchestrator inherits the substrate's fragilities: partition breaks visibility, capacity loss degrades responsiveness, adversarial pressure attacks the routing logic.[4]

The safety-critical systems literature has long recognized the structural problem: a system that monitors itself is not adequately supervised because monitor and monitoree share failure modes. The architectural response, developed in safety kernels and runtime assurance frameworks, is to separate the supervision function into a distinct path that does not share failure surface with the monitoree.[5][6] This paper applies the principle to agentic orchestration.

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1 John Rushby, *Critical System Properties: Survey and Taxonomy*, NASA Contractor Report 189670 (Hampton, VA: NASA Langley Research Center, 1993); refereed companion in Rushby, "Kernels for Safety?," *Safe and Secure Computing Systems*, ed. T. Anderson (Oxford: Blackwell Scientific, 1989).

2 Tim Kelly and Rob Weaver, "The Goal Structuring Notation: A Safety Argument Notation," *DSN 2004 Workshop on Assurance Cases*, IEEE, 2004.

3 Justin H. Kuiper, CISSP, *Operating Model for Agentic Teams* (v1.0-preprint, 2026), zenodo.org/records/19869313.

4 EDO-P1 promotion (in concurrent drafting; yks-build#103).

5 John Rushby, *Critical System Properties: Survey and Taxonomy*, NASA Contractor Report 189670 (Hampton, VA: NASA Langley Research Center, 1993); refereed companion in Rushby, "Kernels for Safety?," *Safe and Secure Computing Systems*, ed. T. Anderson (Oxford: Blackwell Scientific, 1989).

§2 names five failure modes of in-path orchestration on the tactical substrate. §3 specifies the out-of-path AI supervisor. §4 treats supervision as continuous. §5 binds to Skipjack. §6 returns to HGC^3AE^2.

2. FAILURE MODES OF IN-PATH ORCHESTRATION ON THE TACTICAL SUBSTRATE

Partition-induced blindness. When the Orchestrator is partitioned from the substrate it orchestrates, its decisions are made against stale state.

Compromise propagation. A compromised Orchestrator routes compromised work; its in-path position amplifies its compromise.

Supervisor-supervisee coupling. When the supervisor IS the Orchestrator, monitor and monitoree share failure modes; a fault in one is a fault in both.

Latency degradation. In-path supervision contributes to work-flow latency; under capacity-constrained conditions, the contribution may exceed the supervisory benefit.

False-confidence in self-report. An in-path Orchestrator reporting its own health is reporting from the position the report is meant to evaluate; the report carries the bias of its origin.[7]

3. THE OUT-OF-PATH AI SUPERVISOR

The supervisor is an architectural component **distinct from the Orchestrator**, with its own substrate placement, its own state, its own failure surface, and its own decision logic. It observes the Orchestrator's decisions, the substrate's telemetry, and the federation envelope; it emits anomaly signals when observed behavior diverges from expected behavior.

Observation surface. The supervisor reads telemetry from the Orchestrator's routing decisions, the substrate's execution events, the persistence-tier writes (via WAL replay; EDO-P4), and the comprehension instrument's drift signals (MR-P8).[8][9]

Anomaly detection logic. The supervisor compares observed behavior against learned baselines. Anomaly-detection methodology is well-developed in the peer-reviewed literature; the supervisor's contribution is the application to agentic orchestration with envelope-aware thresholds.[10][11]

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6 Patrick Cousot and Radhia Cousot, "Abstract Interpretation: A Unified Lattice Model for Static Analysis of Programs by Construction or Approximation of Fixpoints," *Proceedings of POPL '77*, 1977, https://doi.org/10.1145/512950.512973.

7 Saleema Amershi et al., "Guidelines for Human-AI Interaction," *Proceedings of CHI '19*, 2019, https://doi.org/10.1145/3290605.3300233.

8 Forthcoming MR-P8 *Comprehension Instrument* (v1.0-preprint, in concurrent drafting; yks-build#98).

9 EDO-P4 promotion (in concurrent drafting; yks-build#96).

10 Varun Chandola, Arindam Banerjee, and Vipin Kumar, "Anomaly Detection: A Survey," *ACM Computing Surveys* 41, no. 3 (2009): article 15, https://doi.org/10.1145/1541880.1541882.

11 Klaus Havelund and Grigore Rosu, "Monitoring Programs Using Rewriting," *Proceedings of the 16th IEEE International Conference on Automated Software Engineering (ASE '01)*, 2001, https://doi.org/10.1109/ASE.2001.989813.

Escalation protocol. Anomaly signals escalate to a human operator through the unified observability plane. The supervisor does not take corrective action directly; it surfaces signals. This preserves the H of HGC³AE² — humans retain envelope authority, the supervisor reports.

Supervisor-failure modes. The supervisor itself may fail: partition, capacity loss, compromise. The architecture treats supervisor failure as a separate event from Orchestrator failure; both are signaled to operators; both have independent recovery paths.

4. SUPERVISION AS A CONTINUOUS PROCESS

Three phases recur. **Observation** runs continuously: the supervisor reads telemetry at the cadence the observability plane emits. **Comparison** runs at every Orchestrator decision and at supervisor-internal cadence: observed behavior vs. expected baseline. **Signaling** runs on threshold crossings: anomaly events surface to operators and to the Skipjack reconciliation cycle.

The continuous-process framing distinguishes the supervisor from periodic-audit mechanisms; the supervisor's value is at the time of the anomaly, not at the periodic review.

5. SKIPJACK COUPLING

The supervisor feeds Skipjack's enforcement mechanisms.[12] **Context integrity** is observed (the supervisor sees the Orchestrator's context state through the persistence layer). **Structured routing** is observed (the supervisor sees the Orchestrator's routing decisions). **HITL control points** fire on supervisor-emitted anomalies. **Iterative feedback** closes when supervisor observations feed envelope refinement at the next ceremony.

The supervisor is the operational form of out-of-path runtime assurance for the Skipjack-conformant Orchestrator.

6. IMPLICATIONS AND CONCLUSION

The implications extend to architecture (the supervisor is a first-class component), substrate placement (the supervisor must not share failure surface with the Orchestrator), and operational discipline (anomaly thresholds must be governance-defined, not auto-tuned).

The core claim: **out-of-path AI supervision provides the architectural separation that makes Orchestrator decisions auditable under DDIL conditions; without separation, monitor and monitoree share failure modes and the supervisory commitment is structurally weakened**.

Returned to HGC³AE² letter by letter:

H — Human-governed. Supervisor signals escalate to humans; supervisor does not take corrective action.

G — Curated Context. Supervisor reads curated telemetry — comprehension drift, governance events, persistence consistency.

C¹ — Cybersecurity. Supervisor placement on distinct substrate isolates it from Orchestrator compromise.

12 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

C² — Continuity. Supervisor failure is independent of Orchestrator failure; continuity preserved at both layers.

C³ — Coherence. Supervisor compares observed against expected against a coherent envelope.

A — Agentically Engineered. Anomaly detection is agentic within governance-defined thresholds.

E — Executed. Supervision runs continuously.

The exponent — the closure — is the observation-comparison-signaling loop.

The remaining EDO papers (protocol selection, workload class, classification, accreditation, interoperability, deployment) operate over the supervised orchestration substrate this paper specifies.[13][14]

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13 EDO-P6 promotion (in concurrent drafting; yks-build#105).

14 EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).